

fined to a single hardware token is removed and implemented in software instead, the confidentiality and integrity of the cryptographic mechanism's code and data becomes dependent on the operating system, including both memory protection and file protection. Mandatory security is needed to rigorously ensure the mechanism's integrity and confidentiality. If any external inputs, such as input parameters to a random number generator, are used by the cryptographic mechanism, the input sources and the path between the input sources and the cryptographic mechanism must be protected with mandatory security mechanisms.

System Security: No single technical security solution can provide total system security; a proper balance of security mechanisms must be achieved. Each security mechanism provides specific security functions and should be designed to only provide those functions. It should rely on other mechanisms for support and for required security services. In a secure system, the entire set of mechanisms complement each other so that they collectively provide a complete security package. Systems that fail to achieve this balance will be vulnerable.

A secure operating system is an important and necessary piece to the total system security puzzle, but it is not the only piece. A highly secure operating system would be insufficient without application-specific security built upon it. Certain problems are actually better addressed by security implemented above the operating system. One such example is an electronic commerce system that requires a digital signature on each transaction. A application-space cryptographic mechanism in the transaction system protected by secure operating system features might offer the best system security solution.

No single security mechanism is likely to provide complete protection. Unsolved technical problems, implementation errors and flawed environmental assumptions will result in residual vulnerabilities. As an example, covert channels remain a serious technical challenge for secure operating system designers. These limitations must be understood, and suitable measures must be taken to